

Mutational Signature Analysis

Early-Onset vs. Late-Onset Breast Cancer (TCGA-BRCA)

Project Overview

This analysis pipeline decomposes somatic mutation data from the TCGA-BRCA project into COSMIC v3.3 mutational signatures. The objective is to identify differential mutational processes between early-onset (≤ 45 years) and late-onset (≥ 55 years) breast cancer cohorts.

Methodology

1. Data Acquisition

Somatic mutation data (MAF files) and clinical metadata are retrieved using `TCGAbiolinks`. We specifically target "Masked Somatic Mutations" to ensure high-quality variant calls.

2. Matrix Generation

A 96-trinucleotide substitution matrix is constructed. This matrix counts the frequency of each mutation type (C>A, C>G, C>T, T>A, T>C, T>G) within its 5' and 3' flanking context.

3. Signature Refitting

We apply a non-negative least squares (NNLS) approach to refit the sample matrices against the known COSMIC v3.3 signatures. Key signatures of interest include:

- **SBS3**: Homologous recombination deficiency (HRD).
- **SBS2 & SBS13**: APOBEC cytidine deaminase activity.
- **SBS1 & SBS5**: Endogenous clock-like processes.

Statistical Framework

To compare signature exposures between cohorts, we employ the Mann-Whitney U test. The relative contribution C_{rel} is calculated as:

$$C_{rel} = E_i / \sum E_j$$

Where E_i is the exposure of signature i and the denominator represents the total mutational burden of the sample.